Bihar Engineering University, Patna

B.Tech. 5th Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*

Code: 101503 *Subject: Geotechnical Engineering – I* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Sieving is not practicable for grain sizes smaller than about
(i) 0.075 mm (ii) 0.095 mm
(iii) 0.15 mm (iv) 0.2 mm

(b) The flow index in soils indicates the
(i) ratio of liquid limit to plastic limit (ii) variation of liquid limit
(iii) variation of plastic limit (iv) shear strength variation and water content

(c) A flow line in seepage through a soil medium is defined as the
(i) path of particles of water through a saturated soil mass
(ii) line connecting points of equal head of water
(iii) flow of movement of fine particles of soil
(iv) direction of flow particle

(d) The effective stress on the soil is due to the
(i) external load acting on the soil (ii) weight of the soil particles
(iii) weight of water present in the soil pores (iv) both (i) and (ii)

(e) By means of Boussinesq's stress distribution theory, which one of the following pressure distribution diagrams can be prepared?
(i) Strain isobar
(ii) Vertical pressure distribution on a horizontal plane
(iii) Vertical pressure distribution on an inclined plane
(iv) Pressure distribution diagram on an oblique plane

(f) A soil sample has a porosity of 40 percent. The specific gravity of solids is 2.70. what is void ratio ?
(i) 0.467 (ii) 0.567
(iii) 0.667 (iv) 0.743

(g) A line showing the dry density as a function of water content for soil containing no air voids, is called
(i) saturation line (ii) zero air void line
(iii) liquid limit line (iv) none of these

(h) The critical gradient of a soil increases
(i) with increase in void ratio (ii) with decrease in void ratio
(iii) with decrease in specific gravity (iv) none of these

(i) Consistency is term used to indicate
(i) the quantitative analysis of soils (ii) the degree of firmness of cohesive soils
(iii) the fineness of non-cohesive soils (iv) the fineness of clay soils

(j) Which one of the following does not affect the permeability of soils?
(i) void ratio (ii) soil strength
(iii) grain size (iv) temperature

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Q.3 The soil profile in a particular site consists of 7 m thick sandy layer overlain by a layer of clay. The water table is at 1 m below the ground surface. Above the water table, the sand is saturated with capillary moisture. The dry unit weight of sand is 17 kN/m^3 and its saturated unit weight is 20 kN/m^3. Plot the total stress, neutral stress and effective stress with depth up to a depth of 7 m. [14]

Q.4 An embankment is to be constructed using sandy clay compacted to dry unit weight of 18 kN/m^3. The sandy clay has to be transported to the site from a borrow pit. The bulk unit weight of the sandy clay in the borrow pit is 16 kN/m^3 and its natural water content is 12%. Calculate the volume of sandy clay from the borrow pit required for 1 cubic metre of finished embankment. Assume that the soil swells by 10% due to excavation and during transportation. You can take $G_s = 2.7$. [14]

Q.4 (a) A 1.2 m thick embankment of loose sand is to be compacted using a Vibratory Roller. If the void ratio decreases from 1.2 to 0.8 due to compaction, calculate the final thickness of the embankment. [9]

(b) What is the effect of surcharge and the capillary action on the effective stress? Explain briefly. [5]

Q.5 (a) Two columns A and B are situated 6 m apart. Column A transfers a load of 500 kN and column B, a load of 250 kN. Determine the resultant vertical stress on horizontal plane 20 m below the ground surface at points vertically below the points A and B. [10]

(b) Derive an expression for bulk density in terms of its water content, void ratio, specific gravity of solids and density of water. [4]

Q.6 (a) A sample of clay has a void ratio of 0.70 in the undisturbed state and of 0.50 in a remoulded state. If the specific gravity of solids is 2.65, determine the shrinkage limit in each case. [8]

(b) What is the difference between the classification based on particle size and textural classification? Discuss the limitations of the two systems. [6]

Q.7 (a) Distinguish between discharge velocity and seepage velocity in the case of flow of water through soils. [6]

(b) A soil sample 90 mm high and 6000 mm^2 in cross-section was subjected to a falling head permeability test. The head fell from 500 to 300 mm in 1500 seconds. The permeability of the soil was 2.4×10^{-3} mm/s. determine the diameter of the standpipe. [8]

Q.8 (a) What are the different methods of compaction adopted in the field? How would you select the type of roller to be used? [10]

(b) Atterberg limit tests were carried out on a soil sample, with the results LL as 40% and PL as 25%. Classify the soil according to Unified Classification system and the Indian standard classification system. [4]

Q.9 Write short notes on *any four* of the following: [$3\frac{1}{2} \times 4 = 14$]

(a) Thixotropy
(b) Determination of Unit weight by sand-replacement method
(c) Factors affecting permeability
(d) Quick sand condition
(e) Factors affecting compaction
(f) Newmark's Influence chart

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